Formula 1 Data Analysis

Technological evolution across 70+ years of racing data

[Your Name] · [Course / Class] · [Date]

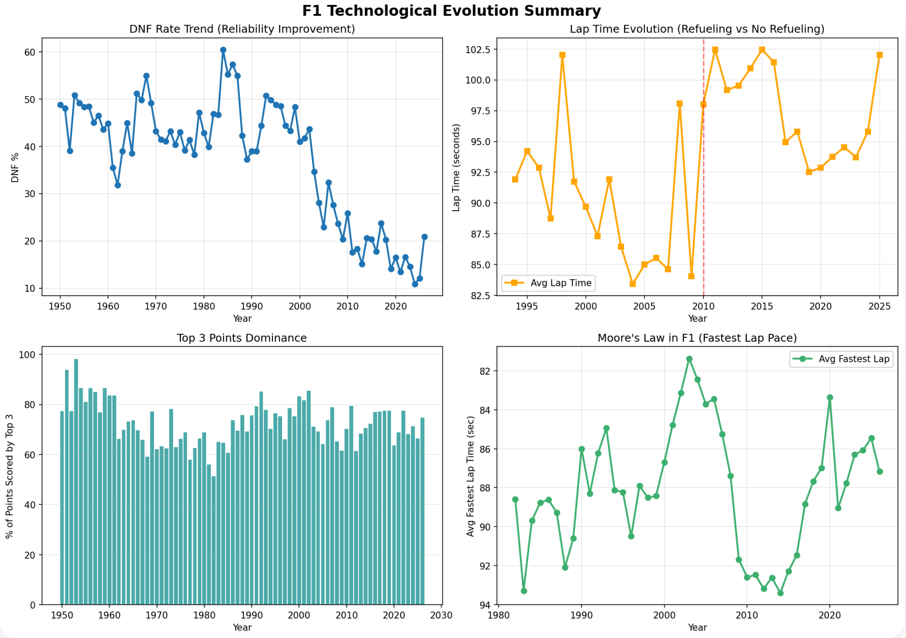
Data: Kaggle (Ergast) F1 dataset · Python · DuckDB · matplotlib

Agenda

1. The dataset & method
2. Team dominance over time
3. Best constructors (points per season)
4. Teammate head-to-head duels
5. Reliability — the DNF story
6. Outright pace — a "Moore's law" check
7. Fuel load impact on lap time
8. Key findings & limitations

1 · The Dataset & Method

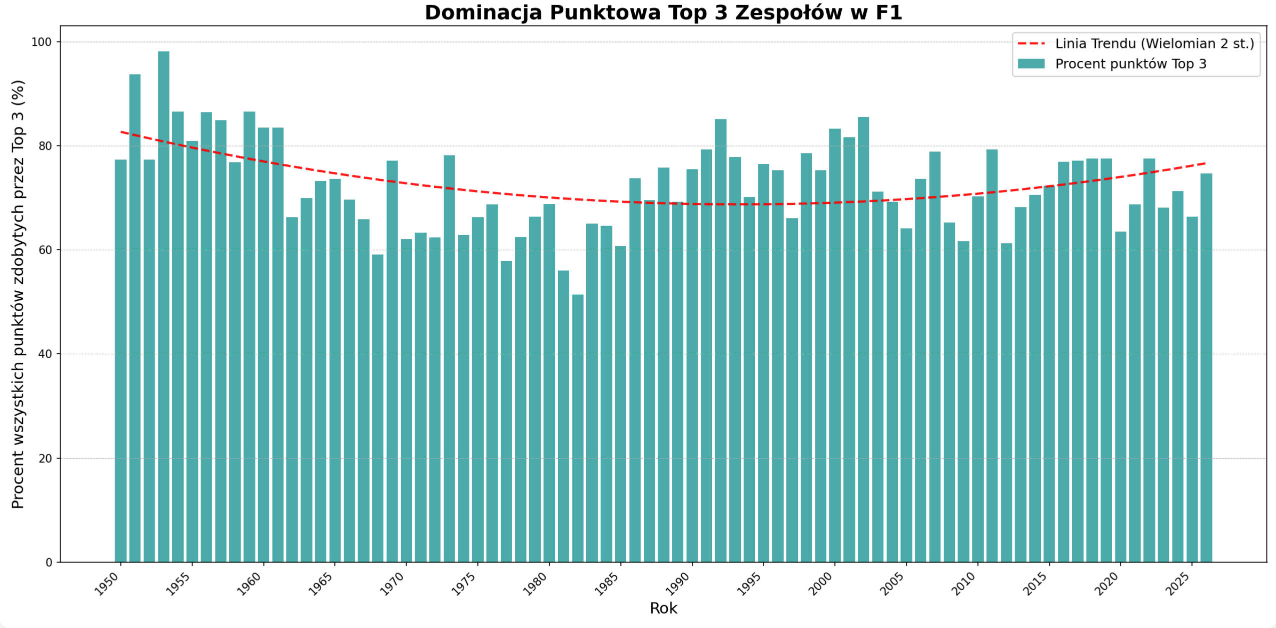
Analysis Overview



Source: Kaggle/Ergast F1 data (1950–2026) in DuckDB. Four headline signals: reliability, pace, top-3 dominance, and the 2010 refuelling-ban shift.

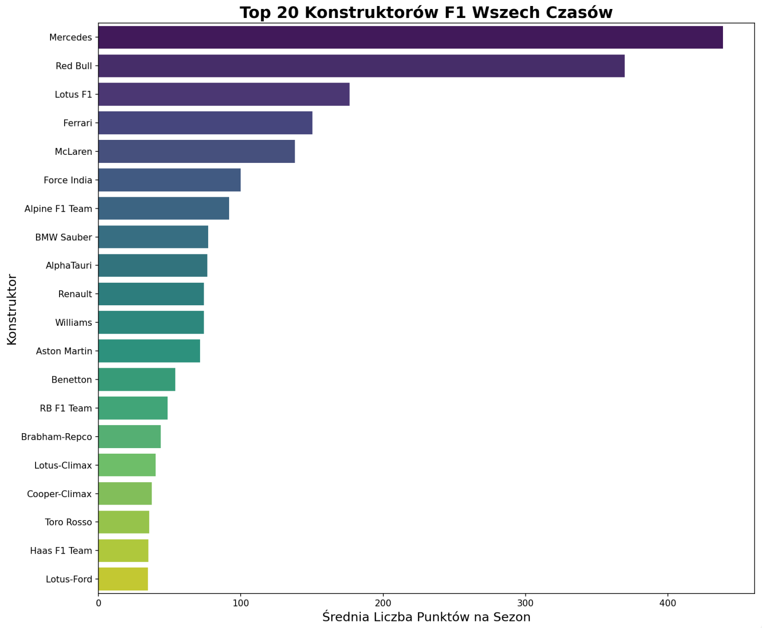
2 · Dominance & Performance

Top-3 Dominance Over Time



Takeaway: The top-3 teams take ~70–80% of all points every season — concentration peaked in the 1950s (~98% in 1952), loosened by the early 1980s (~51% low), and is tightening again today.

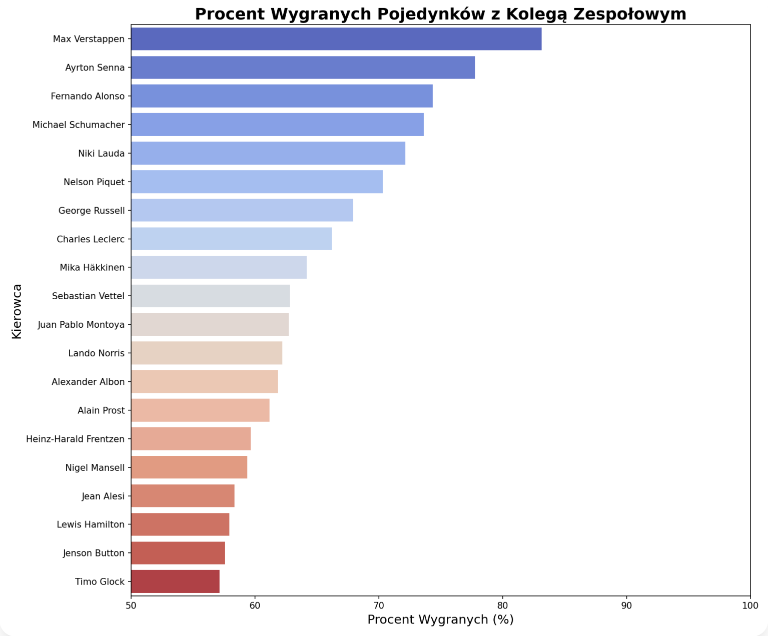
Top Constructors — Points per Season



Takeaway — read with care: Mercedes (~440) and Red Bull (~370) lead average points/season, but this metric rewards the modern high-points era, so it flatters recent teams over Ferrari (~150) and other long-history names.

3 · Driver Battles

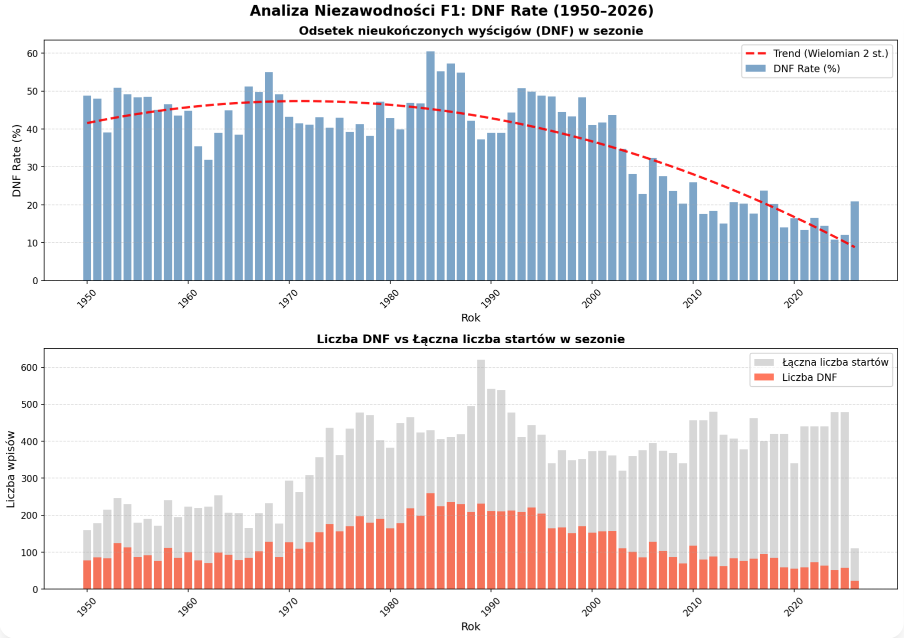
Teammate Duels



Takeaway: Verstappen tops same-car head-to-heads (~83%); Hamilton sits low (~58%) — but the metric counts race finishes only and doesn't adjust for how strong each teammate was.

4 · Reliability

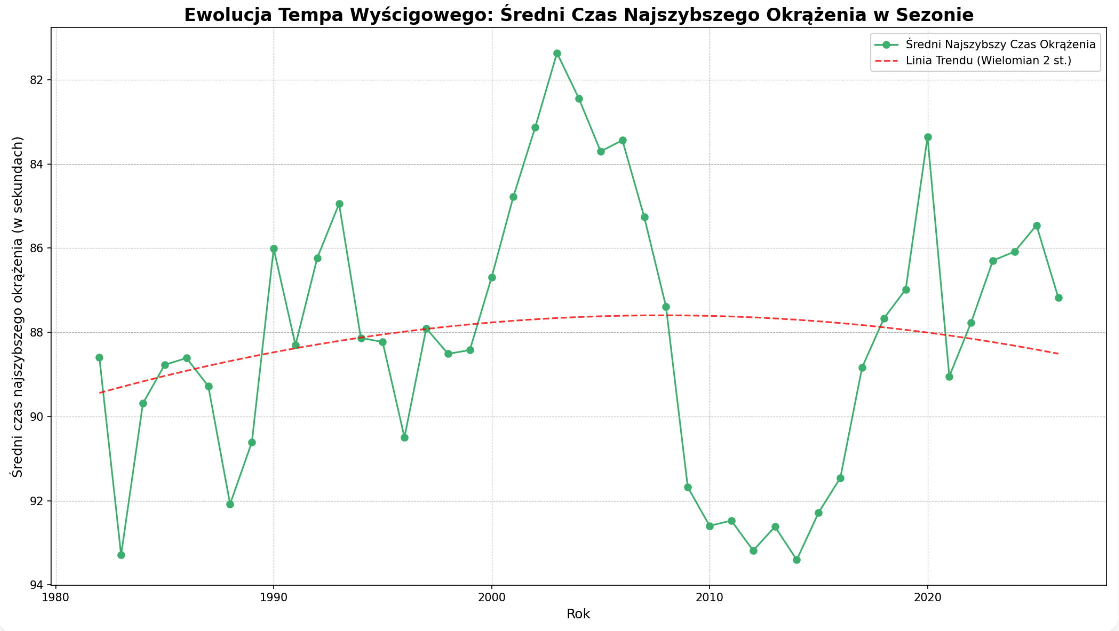
DNF & Reliability Trends



Takeaway: Season DNF rate fell from a ~47% peak (late 1960s) to roughly 10–15% today — finishing went from a coin-flip to the norm. This is the clearest progress story in the data.

5 · Performance Evolution

"Moore's Law" of Fastest Laps?

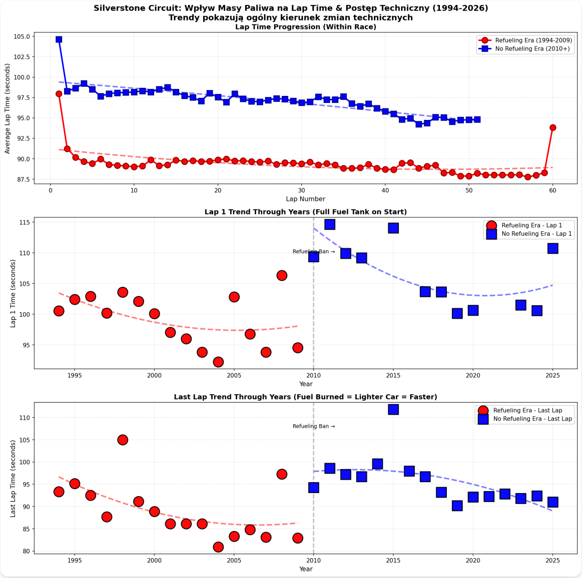


Takeaway: Pace did **not** improve in a straight line — fastest laps peaked in the early 2000s, regressed through the 2009 & 2014 rule changes, then recovered. The "law" only half holds.

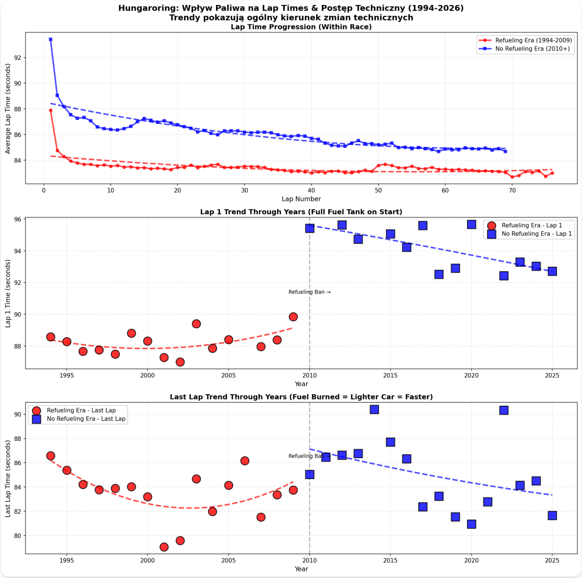
6 · Strategy & Conditions

Fuel Load Impact on Lap Time

As fuel burns off the car gets lighter and faster — the same effect shows on both circuits, and it’s larger in the heavier no-refuelling era (2010+).



Silverstone (most-raced circuit)



Hungaroring (2nd historic circuit)

Key Findings

What the Data Tells Us

- 🔧 **Reliability is the clearest win** — season DNF rate fell roughly 3–4× since the 1960s
- ⛽ **Fuel weight has a measurable, repeatable cost** — visible across circuits and eras
- 🏆 **Dominance is cyclical, not one-way** — top-3 share dipped mid-history, rising again now
- ⚖️ **"Best vs teammate" depends on the metric** — finish-only, opponent strength unweighted
- 📈 **Outright pace is non-monotonic** — regulations reset the curve in 2009 and 2014

Limitations: points-per-season favours the modern era; fastest-lap mixes circuits. This is descriptive analysis — a predictive 2026 model is the intended next step.

Thank You

Questions?

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